Review Sheet 1: Process Analysis and Improvement[1]

Suggested Answers

1. Instant Dolls Co.*

Instant Dolls specializes in manufacturing dolls in a make to order basis. Their business process consists on the following activities, which must be conducted in the sequence described below:

Activity 1: Get the specifications of the doll from the customer, get the payment and submit the order to the production line. (8 minutes)

Activity 2: Assemble the main body of the doll (10 minutes)

Activity 3: Add hair and paint. (12 minutes)

Activity 4: Dress the doll (7 minutes)

Activity 5: Package the doll (3 minutes)

(The time required for each activity is shown in parenthesis)

The whole process is conducted by three workers. The assignment of tasks to workers is the following:

W1 does Activity 1.

W2 does Activities 2 and 3.

W3 does Activities 4 and 5.

a) What is the bottleneck of the process?

We expect W2 to be the bottleneck. This is intuitive because the two activities this worker does, take much longer than activities done by W1 and W3. Recall that the step with the longest processing time or lowest capacity is the bottleneck.

b) How much time will it take to produce 250 units? (assume that the process starts with an empty production line)

*It should be noted that the process starts empty which makes a difference as we detail below. Ignoring this fact and assuming that the process does not start empty, we can simply compute the process time by noting that the time needed to do one order by the bottleneck is 22 mins and thus the time for 250 orders is: 250 *22 mins= 5500.*

Now take the situation when the process starts empty.

When the first order gets to the production, W1 spends 8 mins and passes it to W2 who spends 22 mins and passes it to W3 that spends 10 mins on it and the first product will be out. So the first product takes 8+22+10= 40 mins. Now take the second order. When W1 gets done with the

[1] Problems are rated for difficulty: * = Relatively easy, for first review of concepts; ** = Challenging, assumes good understanding of the concepts; *** = Advanced, assumes full understanding of the concepts and creativity in application.

*first order and passes it to W2, she picks up the second order and works on it which takes only 8 mins. Therefore, when W2 is done with the first one, she will see an order already waiting for her and so she will immediately start working on it (so she does not need to wait to start working). She will take of course 22 mins to finish the work. W3 finishes the first order in 10 mins but then has to wait for W2 to finish the second order, i.e. she has to wait for 22-10=12 mins to get the second order. As soon as she gets it, she will do the job in 10 mins and complete the order. Therefore, the production line spent 22 mins to complete the second order. The same logic applies the remaining orders. We conclude that the total time to be spent on 250 orders is: 40+22*249=5518 mins.*

c) What is the utilization of worker 3?

The input rate should be given to calculate the utilization rates. If not, we usually assume that plants should be operated with the bottleneck utilized 100%. If the bottleneck operates at capacity, then the process produces 60/22=2.73 units/h. This is because it takes 22 mins for W2 (the bottleneck) to complete one order. Therefore, the utilization rate of W3 is: 2.73 units/h /6 units/h = 0.46.

d) What is the average labor utilization of the workers? Assume the process operates at its capacity and there are no empty system effects.

We can use two alternatives for this problem:

Alternative 1:

Flow rate=1/22 units/mins. Therefore:

W1 utilization= (1/22 units/min) / (1/8 units/min)= 8/22= 0.36

W2 utilization=(1/22 units/min) / (1/22 units/min) = 22/22 =1

W3 utilization=(1/22 units/min) / (1/10 units/min) =10/22= 0.45

Therefore, the average utilization is: (0.36+1+0.45)/3= 0.60.

Alternative 2:

Average labor utilization=labor content / (labor content + Sum of idle times across workers)

Labor content = sum of the activity times involving direct labor=8+22+10=40

Idle time for a single worker=cycle time – activity time of the single worker

Because cycle time=1/flow rate=22 mins, then:

Idle time for W1= 22-8=14 mins,

Idle time for W2= 22-22=0 mins

Idle time for W3=22-10=10 mins.

Sum of the idle times= 26 mins

Substituting these in the formula for average labor utilization we have:

Ave. labor utilization= 40 /(40+26)=0.6 .

e) What are the direct labor costs for one doll? Assume a wage rate of $15 per hour.

As calculated in Part b, the flow rate for the process is 1/22 units/min or 60/22 units/hour. It follows that:

*Cost of direct labor= total wages per hour/flow rate per hour= 3 * $ 15 per h / 60/22 units per hour= 16.5 per doll*

To increase the production rate, Instant Dolls is considering two alternatives:

Alternative I: To hire a new worker to help any of the workers without changing the tasks performed by each worker.

Alternative II: To redesign the assignment of tasks to workers. For this, the company is evaluating to re-assign Step 2 from W2 to W1.

f) Suppose that Alternative I is chosen. Where would you assign the new employee?

As seen with the calculations in Part a, W2 has the highest utilization and so is the bottleneck. So I would assign the worker to W2.

g) What would be the direct labor of one doll under the solution depicted above? Assume there is sufficient demand (system operates at its capacity) and there are no empty system effects.

In this case, W2 which now consists two workers completes the work on each doll in 11 mins. It follows that the new flow rate is: 1/11 units/min or 60/11 units/hour.

Using the same approach as in Part e, we have:

*Direct labor cost per doll= 4*15/ (60/11)= $ 11 .*

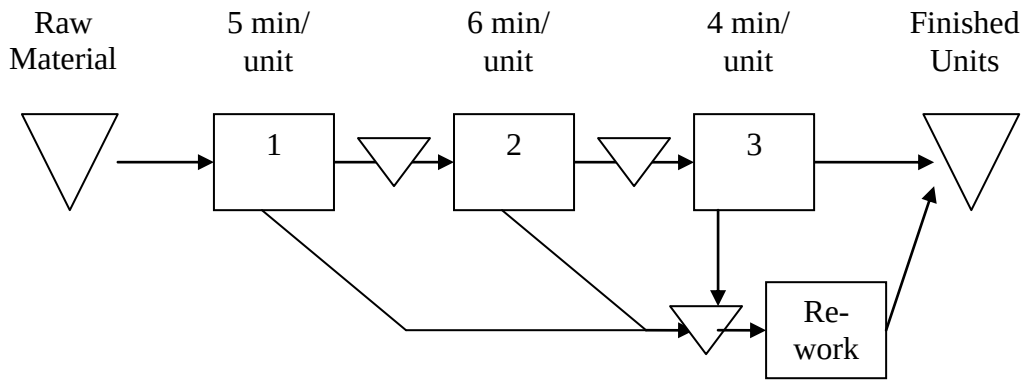
h) What would be the direct labor costs for Alternative II?

It is easy to see that the new flow rate is: 1/18 units/min or 60/18 units/hour. Following the same approach as in Part e, we have:

*Direct labot cost per doll= 3*15 / (60/18)= $ 13.5 .*

2. Ceramics Production**

Will Clintoon manages a glass ceramics production process that has three stages. Each stage operates at the rate that the operator at each stage can work. Squares represent process steps, and triangles represent the potential for work in process inventory.



The process is experiencing serious quality problems related to management providing insufficient training for workers. Approximately 20% of the parts going through stage 1 are badly processed by the operator. Instead of scrapping the unit, it is sent to a highly skilled rework operator that can correct the mistake and finish the unit completely in 15 minutes.

The same problem occurs at stage 2 where 10% of the parts are poorly processed, and require 10 minutes of rework. Stage 3 also has a 10% defect rate, each defect requires 5 minutes of the rework operators time.

(i) What is the utilization of stage 2 if work is released into the system at a rate of 5 units per hour?

The first step in this problem will be to convert all time units onto the same base. One can use the base one feels comfortable with – let us use units per hour.

Then, Capacity of stage 1 = 12 units per hour; Capacity of stage 2 = 10 units per hour; Capacity of stage 3 = 15 units per hour.

Units coming to stage 1 can either go to stage 2 (if they are correctly processed) or to the rework operator (if they are badly processed). If the input rate to stage 1 is $\lambda_1 = 5$ units/hour, then 20% of this input goes to rework operator goes to rework operator. Thus, only 80% of this input reaches stage 2.

Therefore, stage 2 sees units coming at the rate of:

$\lambda_2 = \lambda_1 * 0.8 = 4$ *units/hour*

The capacity of stage 2 is 10 units per hour, so the utilization at stage 2 is

$\rho_2 = 4$ *unit per hour /10 units per hour= 0.4 (40% busy)*

(ii) Where in the process is the bottleneck (i.e., which resource is most highly utilized under any sustainable work release rate)? Why?

We check the utilization at each of the stages. For stage 1 , the process is just like as in (i).

$\rho_1 = 5$ *units per hour/12 units per hour = 0.4167 (41.67% busy)*

The arrival rate of work to stage 3 is $\lambda_3 = \lambda_2 * 0.9 = 4 * 0.9 = 3.6$ *units per hour*

$\rho_3 = 3.6$ *units per hour / 15 units per hour = 0.24 (24% busy)*

For the rework stage, the task is trickier since three types of work might arrive that have three different processing times (one for the rework from each of the three different stages). We also note that the rework stage has different capacities for dealing with each type of flow.

The highly skilled rework operator can correct a defect coming from stage 1 at a rate of 4 units per hour. Similarly, he can correct defects coming from stage 2 at a rate of 6 units per hour and the defects coming from stage 3 at a rate of 12 units per hour.

We therefore compute the amount of utilization at rework caused by defective flows arriving from each individual stage.

The utilization of rework that is due to the 20% defects from stage 1 is:

$\rho_{R1} = 5$ units per hour * 0.2 / 4 units per hour = 0.25

The utilization of rework because of rework from stage 2 is:

$\rho_{R2} =$ 4 units per hour * 0.1 / 6 units per hour = 0.0666

The utilization of rework because of rework from stage 3 is:

$\rho_{R2} =$ 3.6 units per hour * 0.1 / 12 units per hour = 0.03

The overall utilization at Rework is therefore

$\rho_R = \rho_{R1} + \rho_{R2} + \rho_{R3} = 0.25 + 0.066 + 0.03 = 0.3466$ *(34.66 % busy)*

Stage 1 is the most heavily utilized and is currently the bottleneck. Stage 1 is busy 41.67% of the time. The other stages are less busy (40%, 24%, or 34.66% busy).

(iii) After a training program is implemented, the defect rate at stage 1 is reduced to 10% and the defect rates at stage 2 and stage 3 are both reduced to 5%. What is the maximum process capacity after these changes?

Improving the defect rates changes the process flows. Less flows to the rework area. Since it was not a bottleneck before, it will not become a bottleneck now. There is an increase in flow to stages 2 and 3, so these might become bottlenecks now.

With a 10% defect rate at stage 1, stage 2 now sees an increased flow and utilization:

$\lambda_2 = \lambda_1 * 0.9 = 4.5$ *unit per hour*

$\rho_2 = 4.5$ *unit per hour / 10 units per hour = 0.45 (45% busy)*

Stage 3 also sees an increased utilization, but since it sees fewer jobs than stage 2, and the work per unit is less at stage 3 than at stage 2, stage 3 will not be a bottleneck either.

Stage 1 is still seeing 5 units/hour, with capacity of processing 12 units per hour, for a 5/12=41.66% utilization. The quality improvements therefore changed the bottleneck from stage 1 to stage 2.

The theoretical maximum process capacity occurs when the bottleneck is 100% utilized. So the maximum arrival rate to stage 2 can be computed by seeing which rate corresponds to 100% utilization. In other words, we solve:

$\rho_2 = \lambda_{2,MAX}$ /10 units per hour = 1.00 (100% busy)

So the maximum input capacity at stage 2 is
$\lambda_{2,MAX}$ = 10 units per hour

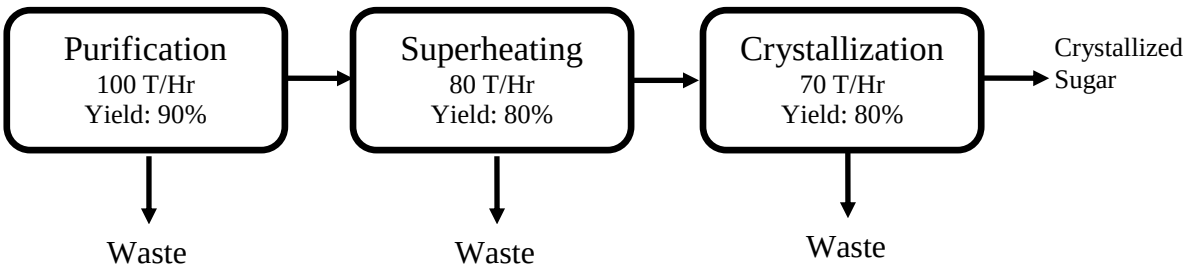
To obtain a maximum capacity of 10 units per hour at stage 2, a higher rate of input to stage 1 is required, since 10% of the units processed at stage 1 go to Rework rather than stage 2. Since $\lambda_2 = \lambda_1 * 0.9$, we have

$\lambda_{1,MAX} = \lambda_{2,MAX} / 0.9 = 10 / 0.9 = 11.11$ units per hour

Of the 11.11 units per hour, 1.11 pass through rework instead of stage 2, but end up exiting the system eventually. The overall maximum capacity is 11.11 units per hour.

3. A Sweet Job

In an unexpected twist of events, you recently inherited a sugar plant from a distant relative. You are looking forward to apply your extensive knowledge of POM principles to this plant.
On your first visit to the sugar producing facility, you learn the following. Sugar production has basically three steps- Purification of Molasses followed by Superheating of the purified molasses and finally crystallization. Your facility has three extremely reliable continuous processing machines for the three process steps. To ensure the freshness of sugar, at no point in the process, can one build work in process inventories. The performance of the individual processing machines is described below.
Step 1, Purification: Capacity- 100 Tons/hour (T/hr) Yield- 90% (in other words, if 100 Tons of input is introduced in the machine, only 90 tons is available for the next step)
Step 2, Superheating: Capacity- 80 Tons/hour. Yield -80%
Step 3, Crystallization: Capacity- 70 Tons/hour. Yield- 80%
The process is illustrated below:



a) If 50 T/hr of molasses are introduced into the purification stage, what would be the utilization of each of the stages? What stage is the bottleneck? What is the capacity of this plant (in Tons of crystallized sugar per hour)? *

Suggested Solution:

Here is a table of flows:

Stage	*Purification*		*Superheating*		*Crystallization*	
	In	*Out*	*In*	*Out*	*In*	*Out*

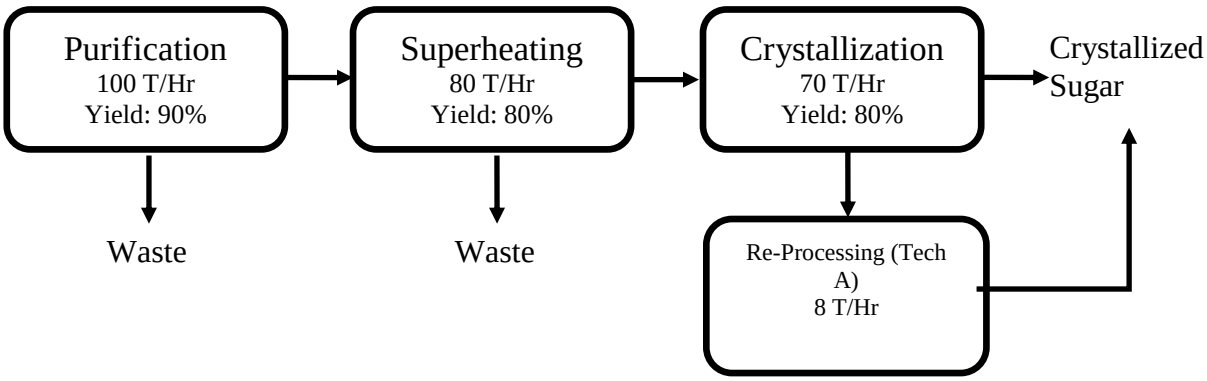
Capacity	*100*		*80*		*70*	
Yield	*90%*		*80%*		*80%*	
A.	*50*	*45*	*45*	*36*	*36*	*28.8*
Utilization	***0.5***		***0.5625***		***0.514286***	

Super-heating is the bottleneck stage, since it has the highest utilization at the sustainable production rate.

To compute the capacity of the plant, the bottleneck would be fully utilized. So the input to the superheating stage must be 80 T/hr. Because superheating produces only 80% good output, then crystallization receives 80T/hr x 0.8 = 64 T/hr input. This stage also has 20% waste, so the final production would be 64 T/hr x 0.8 = 51.2 T/hr, which is the production line capacity.

After analyzing the plant, you feel a little frustrated by the high yield losses. You contact your classmate from INSEAD, who is now the leading process improvement consultant for the sugar industry (What a sweet job!). He points out that there are a couple of new technologies available that can re-process the losses from the crystallization stage. This could increase the total capacity of the plant.

Technology A can take the losses from the crystallization stage and convert all the waste to crystallized sugar (the final product). (For efficient operation of the re-processing stage, it must operate on all the waste from the crystallization stage. In other words, ***all*** the waste that comes from the crystallization stage must be introduced into the Re-processing stage.) The modified Process diagram is illustrated below.



b) *If you buy a Technology-A Re-processing machine with a capacity of 8 T/Hr, what is the new capacity of the plant? What is the minimum capacity of the new machine that would not change the bottleneck from the one you found in Part A above? **

Suggested Solution: The Re-processing stage becomes the new bottleneck.

Stage	*Purification*		*Superheating*		*Crystallization*		*Re-Processing*	
	In	*Out*	*In*	*Out*	*In*	*Out*	*In*	*Out*
Capacity	*100*		*80*		*70*		*8*	
Yield	*90%*		*80%*		*80%*		*100%*	
A.	*50*	*45*	*45*	*36*	*36*	*28.8*	*7.2*	*7.2*

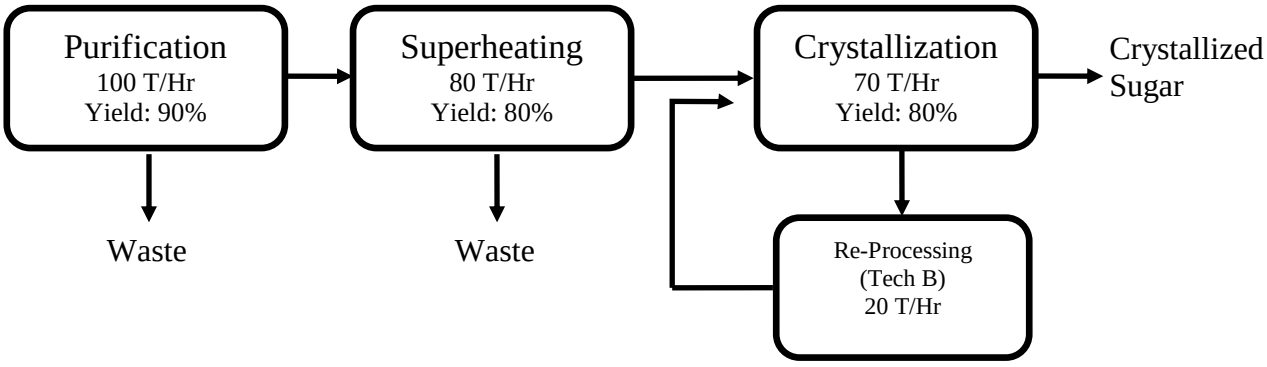
Utilization	*0.5*	*0.5625*	*0.514286*	*0.9*

To get a 100% utilization of the re-processing stage, we need to introduce 40 T/hr in the crystallization and get a 40 T/hr × 0.2 = 8 T/hr waste to keep the bottleneck busy. Crystallization then produces 32 T/hr and Re-processing finishes the 8 T/hr and so the capacity of the process is 40 T/hr.

To ensure there is no new bottleneck, the re-processing stage should not be utilized any more than 56.25% for 50 T of input, which is the utilization of superheating stage for 50 T/hr input. As calculated in the table, for this amount of input into the production line, the amount of input that gets to the re-processing stage is 7.2 T/hr. So the capacity must be such that: 7.2 / Capacity = .5625 or capacity = 7.2/(0.5625) = 12.8 T/hr.

(Note the similarity of this question to Problem 2 in Review Sheet 1).

Technology B on the other hand takes the losses from the crystallization stage and converts them to super-heated liquid, which must be again re-introduced into the crystallization stage. The process diagram is illustrated below



c) ***If you buy a Technology-B Re-processing machine with a capacity of 20 T/Hr, what is the new capacity of the plant? (Like Technology A, efficient operation of Technology B requires that all the waste from the crystallization stage be introduced into the re-processing stage) ******

The only hard thing in this case is to note that some of the output gets recycled. Say at the crystallization stage, X units come from superheating and Y units from the reprocessing stage. Then, we can write a flow equation as Y=0.2(X+Y). Or Y=X/4. Or total flow into Crystallization is 5/4 times the output of the superheating stage.

On computing the flows, one will notice that the crystallization stage is now the bottleneck.

Stage	***Purification***		***Superheating***		***Crystallization***		***Re-Processing***	
	In	*Out*	*In*	*Out*	*In*	*Out*	*In*	*Out*
Capacity	*100*		*80*		*70*		*20*	
Yield	*90%*		*80%*		*80%*		*100%*	
A.	*50*	*45*	*45*	*36*	*45*	*36*	*9*	*9*
Utilization	*0.50*		*0.5625*		*0.6429*		*0.45*	

As for the previous case, to compute the capacity of the process the bottleneck would need to be fully utilized. The calculation then shows that the capacity of this system is 56 T/Hr.

4. Sport Obermeyer *

Sport Obermeyer manufactures parkas using an assembly line. It owns two factories, one in Mexico and the other in China. The workers in the Chinese factory are multi-skilled. The workers in Mexico are relatively less skilled and more specialized. Sport Obermeyer uses an assembly line with twenty workers in China and one with 40 workers in Mexico. The capacities are the same in the two assembly lines. Based on this information you can conclude that relative to the factory in China (explain to get credit):

a. The Mexican factory makes more parkas per hour but takes less time to make parkas.
b. The Mexican factory makes more parkas per hour but takes longer to make parkas.
c. The Mexican factory makes the same number of parkas per hour but takes longer to make parkas.
d. The Mexican factory makes the same number of parkas per hour but takes the same time to make parkas.
e. None of the above

(Note: Time to make a parka means the time from beginning to end of the assembly line. Assume that the transfer lot size is one, that is when a worker finishes one operation the garment is passed to the next worker for the next operation.)

We use Little's Law to solve the problem.
The capacities in the two factories are the same. Therefore the throughput is the same, assuming the factories are working at full capacity.
Inventory is the work in progress. There are 40 parkas on the assembly line in Mexico at any given point (we are assuming no idle workers). There are just 20 parkas on the assembly line in China at any given point.
Then it must be that the flow time in China is lower than the one in Mexico.
The idea is that more unskilled workers (more steps in the manufacturing) lead to more inventory lying around. More work in progress in one factory, but same capacity translates into longer flow time in that factory.
Answer: c